SonoChroma AI:
Deep Learning-Based Colour Restoration for Precision Liver Cyst Diagnostics.

2026 Harvard Health Systems Innovation Hackathon | Singapore Hub

- **Engineering Lead:** PhD Candidate, NTU MAE (Advanced Materials & DIW)
- **Clinical Strategy:** Medical Consultant & Health Services Researcher
- **AI/ML Leads:** Researchers in Computational Sciences

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"The Imaging is Grayscale, but Cyst is NOT."

Liver cyst puncture can be advisable or must AVOID

17.6/100,000

Liver Cyst Prevalence

of Asians affected by different liver cyst

$500+

MRI / CT Cost

accurate but inaccessible and contrast enhancement issue

75-85%

Ultrasound Specificity

cheap and fast yet subjective

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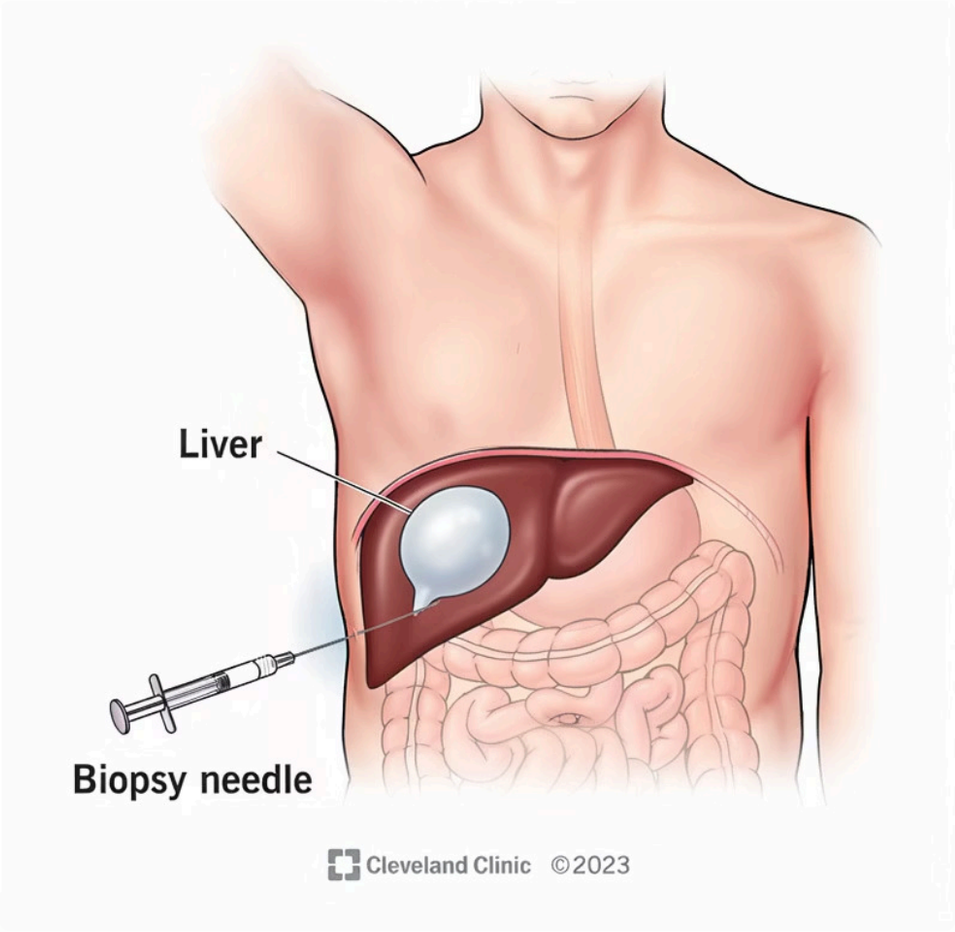
Bridging AI Feature Extraction and Human Visualisation



Deep Learning Pipeline

- A cyst classification based on shape, echogenisity/density, border and size of cyst - these shall be our **learnable** parameters.
- Attention color to indicate **or** avoid liver puncture
- Optimising for **sensitivity & specificity** through repeated iterations (usage)

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Pure Software. Seamless Integration.

Clinic Ultrasound

Use existing machine — no purchase

SaaS Platform

Real-time AI analysis and support

Physician Terminal

Expert-grade diagnostic confidence

Deployed as a **pure SaaS medical device**, SonoChroma AI integrates with any existing real-time ultrasound machine — no new hardware procurement required. It elevates primary care physicians to specialist-level diagnostic confidence, significantly reducing the need for invasive biopsies.

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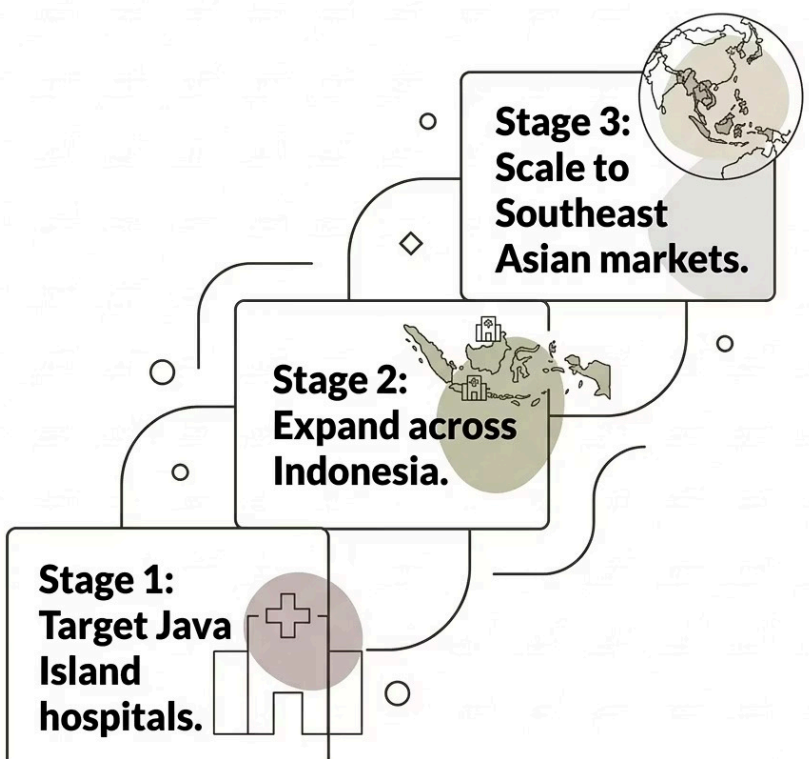
B2B SaaS — High Margin, Borderless Scale

AI ultrasound market reached **USD 1.82 billion in 2025** and is expected to grow to **USD 13.56 billion by 2033**

Revenue Model

- **Pay-per-scan** — low barrier to adoption
- **Annual subscription** — predictable recurring revenue
- Zero hardware dependency = **maximum scalability**

Go-to-Market Roadmap



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Cross-Disciplinary. Full-Cycle. Ready to Deploy.

System Architect

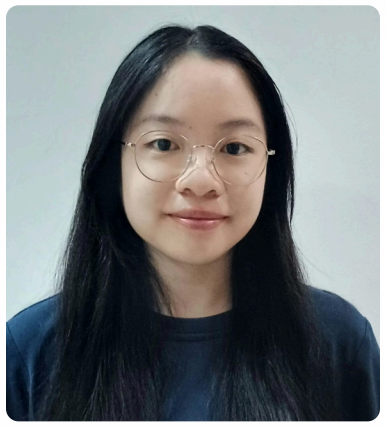
Materials & Systems Engineering — hardware-software integration

Medical Doctor

Clinical validation, IRB compliance & physician workflow design

AI / ML Architect

Deep learning pipeline, fine-tuned model(s) for compatibility



Our Ask Today

- **Hospital Data Partnerships** — 1,000 de-identified, annotated ultrasound datasets for training
- **Seed Funding** — for field testing in Indonesia

> "SonoChroma AI — making expert-grade liver diagnosis as affordable and accessible as a standard ultrasound."

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